

Dr. John D. Kirwan

Computational Biologist · Bioinformatics & Machine Learning

San Cristóbal de la Laguna, Spain • john@jkirwan.org • +34 639981732 • jkirwan.org



KEY SKILLS

- Python
- R
- PyTorch
- Machine Learning
- Deep Learning
- Bayesian Statistics
- Protein Structure Prediction
- Structural Bioinformatics
- Metagenomics
- Docker

PROFILES

GitHub: JohnKirwan
LinkedIn: johndkirwan
ORCID:
0000-0001-5537-3574

LANGUAGES

English (Native)
Spanish (B2)
Swedish (B2)

PROFILE

Computational biologist and data scientist pairing rigorous statistical modelling — Bayesian inference and deep learning — with hands-on ML engineering and reproducible pipelines. Nearly four years at the interface of biotech and AI in an industry research setting, spanning biomolecular structure prediction, generative molecular design, and metagenomic analysis, and including a contribution to a NeurIPS 2025 publication on uncertainty quantification. Now focusing on structure-informed metagenomics, functional annotation, and antimicrobial-resistance research — applying structural and evolutionary insight to large-scale biological data.

SELECTED HIGHLIGHTS

- Computational biologist with nearly four years at the AI-biotech interface in industry research
- Experience spanning structure prediction, binding-affinity modelling, metagenomics, and uncertainty quantification
- Hands-on ML engineering with PyTorch, Docker, AWS, Azure DevOps, and GPU workflows
- Research background grounded in Bayesian modelling, quantitative biology, and reproducible analysis

RECENT EXPERIENCE

Bioinformatics Scientist *Arquimea Research Centre, La Laguna, Spain* (Jul 2022-Mar 2026)
Worked at the interface of the AI and biotech teams on structure-based drug discovery, generative molecular design, and applied machine learning — the link between the computational tools and the biology. Contributing author on a NeurIPS 2025 publication on uncertainty quantification in deep regression.

- Built a modular synthetic-data-generation workflow orchestrating GNINA, DiffDock and AlphaFold 2; sourced datasets and handled inter-tool format conversion
- Predicted protein structures and binding affinities for protein complexes, including metal-ion and ATP cofactors, using Boltz, AlphaFold, Foldseek and BioEMU
- On the engine team of Uncharted, an internal venture building a multi-objective small-molecule optimisation platform; developed the binding-affinity objective

Postdoctoral Researcher *Stazione Zoologica Anton Dohrn, Naples, Italy* (Oct 2019-Jul 2022)
Quantitative sensory biology within an international Human Frontier Science Program (HFSP) collaboration, with emphasis on statistical modelling and signal processing of behavioural data.

- Analysed experimental data using Bayesian multilevel regression models in R and Stan
- Designed and executed behavioural experiments with automated data acquisition pipelines
- Contributed to interdisciplinary publications spanning biology, optics, and computation

EDUCATION

PhD Sensory Biology *Lund University, Sweden* (May 2013-May 2018) Thesis: ‘Spatial Vision in Diverse Invertebrates’.

MSc Molecular Evolution *University College Dublin, Ireland* (Dec 2008-Jun 2010) Thesis: ‘The Molecular Evolution of Hearing in Mammals’

BSc (Hons) Zoology *University College Dublin, Ireland* (Sep 2004-May 2008)
• First Class Honours (GPA: 4.02/4.2)

ADDITIONAL EXPERIENCE

Project Manager (contract) *Lund University, Lund, Sweden* (Jun 2019-Aug 2019) Short-term research management contract supervising experimental data collection and a masters student, bridging doctoral and postdoctoral positions.

Doctoral Researcher *Lund University, Lund, Sweden* (May 2013-Jun 2018) Computational and statistical analysis of visual systems in invertebrates, combining experimental biology with quantitative modelling.

- Built statistical models using Bayesian multilevel regression in R and Stan; additional analysis in Python and Matlab
- Developed optical and geometric models to estimate spatial resolution theoretically
- Designed novel experimental protocols for quantifying animal visual performance

AWARDS & CERTIFICATIONS

- Pharmaceutical Bioinformatics & Applied Pharmaceutical Bioinformatics — Uppsala University — 2020 Pass with Distinction
- Customising your models with TensorFlow 2 — Coursera / Imperial College London — 2025
- Valohai MLOps Fundamentals Certification — Valohai — 2025
- Applied Software Engineering Fundamentals Specialization — Coursera / IBM — 2023 Covers Git, GitHub, Linux, and Python, and the fundamentals of software engineering.
- Mountain Leader Award — Mountain Leader Training Scotland — 2011 Professional certification to lead hillwalking groups.

PUBLICATIONS

Dancing in the dark: The annelid *Platynereis dumerilii* is re-envisioned for its climactic final night John D. Kirwan, Emelie A. Brodrick, Jacob Bartholin, Morten Bartholin, Julius Friis Petersen, Cameron Hird, Anders Garm *Journal of Experimental Biology*, 2025 doi.org/10.1242/jeb.251047

Uncertainty Quantification for Deep Regression using Contextualised Normalizing Flows Adriel Sosa Marco, John D. Kirwan, Alexia Toumpa, Simos Gerasimou *The Thirty-ninth Annual Conference on Neural Information Processing Systems, 2025* openreview.net/forum?id=ZL7RuWd6QK

Phenotypic response to food availability in sea urchin larvae and impact of light during development and growth Maria Cocurullo, John D. Kirwan, Maria Ina Arnone *Frontiers in Ecology and Evolution*, 2025 www.frontiersin.org/journals/ecology-and-evolution/articles/10.3389/fevo.2025.1548208/full

A Model of Decentralized Vision in the Sea Urchin *Diadema africanum* Tianshu Li, John D. Kirwan, Maria Ina Arnone, Dan-Eric Nilsson, Giancarlo La Camera *iScience*, 2023 www.cell.com/iscience/fulltext/S2589-0042(23)00372-3

Run and Hide: Visual Performance in a Brittle Star Lauren Sumner-Rooney, John D. Kirwan, Carsten Lüter, Esther Ullrich-Lüter *Journal of Experimental Biology*, 2021 journals.biologists.com/jeb/article-pdf/doi/10.1242/jeb.236653/1993087/jeb236653.pdf

Chicken colour discrimination depends on background colour Peter Olsson, Robin D. Johnsson, James J. Foster, John D. Kirwan, Olle Lind, Almut Kelber *Journal of Experimental Biology*, 2020 jeb.biologists.org/content/223/24/jeb209429.full.pdf

Extraocular Vision in a Brittle Star Is Mediated by Chromatophore Movement in Response to Ambient Light Lauren Sumner-Rooney, John D. Kirwan, Elijah Lowe, Esther Ullrich-Lüter *Current Biology*, 2020 www.cell.com/current-biology/abstract/S0960-9822(19)31512-X

A millipede compound eye mediating low-resolution vision John D. Kirwan, Dan-E. Nilsson *Vision Research*, 2019 www.sciencedirect.com/science/article/pii/S0042698919301725

The sea urchin *Diadema africanum* uses low resolution vision to find shelter and deter enemies John D. Kirwan, Michael J. Bok, Jochen Smolka, James J. Foster, José Carlos Hernández, Dan-Eric Nilsson *Journal of Experimental Biology*, 2018 jeb.biologists.org/content/221/14/jeb176271

Low-resolution vision in a velvet worm (*Onychophora*) John D. Kirwan, Josefine Graf, Jochen Smolka, Georg Mayer, Miriam J. Henze, Dan-Eric Nilsson *Journal of Experimental Biology*, 2018 jeb.biologists.org/content/221/11/jeb175802

Spatial vision in diverse invertebrates John D. Kirwan 2018 portal.research.lu.se/portal/en/publications/spatial-vision-in-diverse-invertebrates(dc76fe19-18de-49e6-955a-47d1a084760c).html

A phylomedicine approach to understanding the evolution of auditory sensory perception and disease in mammals John D. Kirwan, Michaël Bekaert, Jennifer M. Commins, Kalina T. J. Davies, Stephen J. Rossiter, Emma C. Teeling *Evolutionary Applications*, 2013 doi.wiley.com/10.1111/eva.12047